

Yidan Xu (she/her)

CONTACT INFORMATION	Department of Statistics 436 West Hall University of Michigan Ann Arbor, Michigan, MI 48109	Webpage: https://edenx.github.io Email: yidanxu@umich.edu Tel: +1 (206)-409-9027 Language: English, Mandarin
RESEARCH INTERESTS	- Causal Inference - Optimal Transport - Generative Models - Bayesian Nonparametrics	
EDUCATION	University of Michigan, Ann Arbor, USA 09/2021-Present. <i>Ph.D. Candidate, Statistics</i> Advisors: Yixin Wang, Xuanlong Nguyen Expected Year of Graduation: Fall 2026 University of Washington, Seattle, USA 09/2019 - 06/2021 <i>M.S., Statistics</i> Graduation commencement speaker, representative of Class 2019 MS graduates. Imperial College London, London, UK 09/2016 - 06/2019 <i>B.Sc., Mathematics with Statistics</i> Dean's List 2019, top 5% of 200 students.	
PUBLICATION	(† co-first author)	
ACCEPTED	Semenova, E.[†], Xu, Y.[†], Howes, A., Rashid, T., Bhatt, S., Mishra, S., & Flaxman, S. (2022). PriorVAE: encoding spatial priors with variational autoencoders for small-area estimation. <i>Journal of the Royal Society Interface</i>, 19(191), 20220094. Zhu, Q.[†], Yang, C.[†], Xu, Y., Wang, H., Zhang, C., & Han, J. (2021). Transfer learning of graph neural networks with ego-graph information maximization. <i>Advances in Neural Information Processing Systems</i>.	
IN PROGRESS	Xu, Y.[†], Wang, Y. & Nguyen, X. Stable causal inference with optimal transport. - Developed covariate adjustment with Optimal Transport map. - Improved on stability of causal estimates under various overlap conditions. - Established asymptotic properties of the proposed estimators. Xu, Y.[†], Wang, Y. & Nguyen, X. Partial identification of causal effect in probability spaces. - Proposed a sensitivity analysis framework with Wasserstein metric. - Derived sensitivity bounds for causal estimands with Wasserstein Distributionally Robust Estimation. - Established asymptotic properties for the partial identification bounds. Xu, Y.[†], Wang, Y. & Nguyen, X. Flow matching for efficient inference in structured graphical models. - Proposed a flow matching objective that incorporates unobserved latent structure. - Employed continuous normalizing flow for both likelihood and posterior classes.	

- Compared with existing work for an array of inference tasks.

Xu, Y.[†], Wang, Y. & Nguyen, X. A distributional approach to mixture model estimation.

- Extended deconvolution to general latent spaces with given likelihood kernels.
- Developed a variational optimization procedure in estimating unobserved discrete latent measure.
- Compared with existing method for various simulation settings.

AWARDS

Mary Lister McCammon Research Fellowship 2019 Summer, ICL
 - Awarded to 14 female undergraduate students in Mathematics and Statistics across UK.

BP Undergraduate Research Opportunities Project Awards 2018 Summer, ICL
 - Awarded to students undertaking UROP project at Imperial College, 1 in every department.

COMPUTING SKILLS

Programming Language
 Python, R, MATLAB

Tool & Framework
 JAX, NumPyro, Stan, INLA, Pytorch, TensorFlow